Numerical Analysis of 1-D Thermal Conduction in a Plasma Slab

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1. Introduction

Nuclear fusion presents considerable potential as a sustainable energy source, offering the possibility of plentiful, low-carbon energy from hydrogen isotopes in seawater. In a fusion reactor, energy is generated when light nuclei, deuterium and tritium in most current designs, merge to create helium and release high-energy neutrons. Reaching and maintaining the extreme conditions needed for fusion, specifically temperatures above 100 million Kelvin, requires confining a plasma: a completely ionized, electrically neutral mixture of ions and electrons. The tokamak is the most advanced method for magnetic confinement fusion, featuring a toroidal design that utilizes both externally applied and internally generated magnetic fields to prevent hot plasma from touching material surfaces (Freidberg, 2007). Two prominent instances of this strategy are ITER, the global experimental reactor being built in France, and the under-development SPARC device created by Commonwealth Fusion Systems, both of which seek to demonstrate net energy gain from fusion processes.

A major milestone for fusion reactors is ignition. Ignition is the condition in which the heating of alpha particles alone sustains the plasma against any energy loss – the energy accumulated in the plasma must balance or exceed the energy lost during transport. Among these losses, thermal conduction is the dominant energy dissipation mechanism in magnetically confined plasmas (Freidberg, 2007). Thermal energy diffuses from the hot plasma core to the colder boundary under the effect of a temperature gradient. This outward heat flow eventually reaches the first wall: the innermost surface of the reactor facing the plasma. The first wall must absorb and dissipate this heat load without melting or structural damage, making its thermal performance a central design constraint in fusion engineering. Candidate materials for first-wall applications include tungsten, which has a high melting point and low sputtering efficiency, and various structural steels used in breeder-reactor shell modules (Zinkle & Was, 2013).

A basic model for studying plasma heat conduction is the one-dimensional plasma sheet, in which the plasma occupies an area $-a \leq x \leq a$ and the temperature changes only in the x direction. Although simplified compared to the complete toroidal geometry, this model captures the essential physics of the power balance problem and provides a practical framework for analyzing how heat diffusion and internal heating affect steady-state temperature characteristics. This approach follows the treatment presented by Freidberg (2007, section 14.1), in which the energy balance in the plate is reduced to a second-order ordinary differential equation. This model is particularly useful for understanding how changes in the transport coefficient and heat source configuration affect the maximum plasma temperature and thus the heat load exerted on the first wall.

1.1 Governing Equations and Assumptions

The steady-state energy balance for a plasma in one dimension is governed by the condition that the divergence of the heat flux equals the volumetric heating source. Following Freidberg (2007, Eq. 14.10), this is expressed as:

$$d/dx\ (n\chi\ dT/dx) + S = 0$$

where n is the plasma number density [m^{-3}], χ is the thermal diffusivity [$m^2\ s^{-1}$], T(x) is the temperature [keV], and S is the volumetric heating rate [$W\ m^{-3}$]. The heat flux q is defined by Fourier's law for plasmas as (Freidberg, 2007, Eq. 14.5):

$$q = -n\chi \, dT/dx$$

This second-order ODE is solved subject to symmetric boundary conditions. By symmetry, the temperature gradient vanishes at the center of the slab, $dT/dx|_{x=0} = 0$, and the temperature is prescribed at the plasma edge, $T(\pm a) = T_w$, where T_w is the wall temperature (taken as $T_w = 0$ in the normalized formulation). These conditions reduce the problem to a standard boundary value problem (BVP) amenable to numerical finite difference methods.

Several simplifying assumptions are adopted in this model. First, the plasma is assumed to be in a steady state, so all time derivatives are neglected. Second, the transport is treated as purely one-dimensional, with no variation in the poloidal or toroidal directions. Third, in the baseline case, the plasma density n and thermal diffusivity χ are treated as uniform constants, which allows an analytical parabolic solution to serve as a verification benchmark (Freidberg, 2007, Eq. 14.61). Fourth, radiation and convective losses are neglected, so the only energy loss mechanism is conduction. The model is then extended to examine non-uniform heating profiles S(x), which cannot generally be solved analytically and require the numerical solver.

1.2 Project Objectives

The objectives of this project are threefold. The primary objective is to develop a finite difference solver in Python for the 1-D plasma energy balance equation and verify its accuracy against the known analytical parabolic solution under uniform heating. The second objective is to use the verified solver to conduct sensitivity studies, examining how the peak plasma temperature responds to variations in thermal diffusivity χ and non-uniform heating source profiles S(x). The third objective is to interpret the numerical results in the context of the fusion power balance problem, drawing physical conclusions about the conditions under which ignition becomes feasible. Together, these goals demonstrate the practical utility of numerical boundary value problem solvers in plasma physics and fusion engineering.

2. Numerical Methods

2.1 Discretization of the Governing Equation

The governing steady-state energy balance equation, under the assumption of constant n and χ, reduces to a second-order linear ODE:

$$d^2T/dx^2 = -S(x) / (n \chi \, k_conv)$$

where $k_conv = 1.602 \times 10^{-16}$ J/keV is the unit conversion factor from joules to kiloelectronvolts. The spatial domain $[-a, a]$ is discretized into N uniformly spaced grid points, yielding a grid spacing $\Delta x = 2a / (N - 1)$. At each interior node i, the second derivative is approximated using the standard second-order central difference formula (Freidberg, 2007; Quintana-Kuether, 2026):

$$d^2T/dx^2 \approx (T_{i-1} - 2T_i + T_{i+1}) / (\Delta x)^2$$

Substituting this approximation into the governing equation at each interior node produces a system of algebraic equations of the form:

$$T_{i-1} - 2T_i + T_{i+1} = -S_i (\Delta x)^2 / (n \chi \, k_conv) \quad \text{for } i = 1, 2, \ldots, N-2$$

This is a tridiagonal system with coefficients $[1, -2, 1]$ on the left-hand side and the discretized source term on the right-hand side. The boundary conditions $T(-a) = T(a) = 0$ are enforced

directly by fixing T_0 = T_{N-1} = 0, which replaces the first and last rows of the matrix with identity conditions. The result is an N×N tridiagonal linear system $A \cdot T = B$, which is solved using NumPy's dense direct solver (numpy.linalg.solve). While a dedicated tridiagonal solver such as the Thomas algorithm would be more computationally efficient, the dense solver is appropriate here given the small problem sizes ($N \leq 320$) used in this study.

2.2 Implementation in Python

The solver is implemented as a Python function solve_plasma_temp(S_profile, N, n, chi, a_val), which accepts a heating source array, grid resolution, and physical parameters as inputs and returns the spatial grid and temperature profile in keV. This function signature allows the same solver to be called across all study cases, verification, convergence analysis, heating profile sensitivity, and diffusivity sensitivity, ensuring consistency and reproducibility.

The tridiagonal coefficient matrix A is constructed by looping over all interior nodes and setting the stencil coefficients. Dirichlet boundary conditions are enforced by setting A[0,0] = A[N-1,N-1] = 1 with corresponding zero right-hand-side entries. The source term vector B is populated by evaluating the discretized right-hand side at each node. The system is then solved with a single call to numpy.linalg.solve, and the resulting temperature vector is returned directly in keV.

2.3 Verification Against the Analytical Solution

To rigorously verify the accuracy of the numerical solver, a non-uniform Gaussian heating profile was utilized, defined as $S(x) = S_0 \exp(-(x/w)^2)$ where the characteristic width was set to $w = 0.4a$. While a simple uniform heating source yields a parabolic temperature profile, central finite difference schemes are mathematically exact for polynomials up to degree two. Therefore, employing a Gaussian source provides a significantly more robust benchmark. It yields a non-polynomial analytical exact solution involving the Error Function (erf), ensuring the solver is truly tested against spatial truncation errors rather than passing by mathematical coincidence.

The exact analytical solution was evaluated and compared against the numerical output using the baseline plasma parameters ($n = 1.0 \times 10^{20}$ m^{-3}, $\chi = 1.0$ m^2/s, $S_0 = 1.0$ MW/m^3, $N = 100$). Figure 1 presents the overlay of the exact analytical solution and the Finite Difference Method (FDM) numerical approximation. The two curves are visually indistinguishable, perfectly tracking the temperature gradients and peaking precisely at the same core temperature. This exact alignment confirms the correctness of the coefficient matrix assembly, the enforcement of Dirichlet boundary conditions, and the joule-to-kiloelectronvolt unit conversions within the implementation.

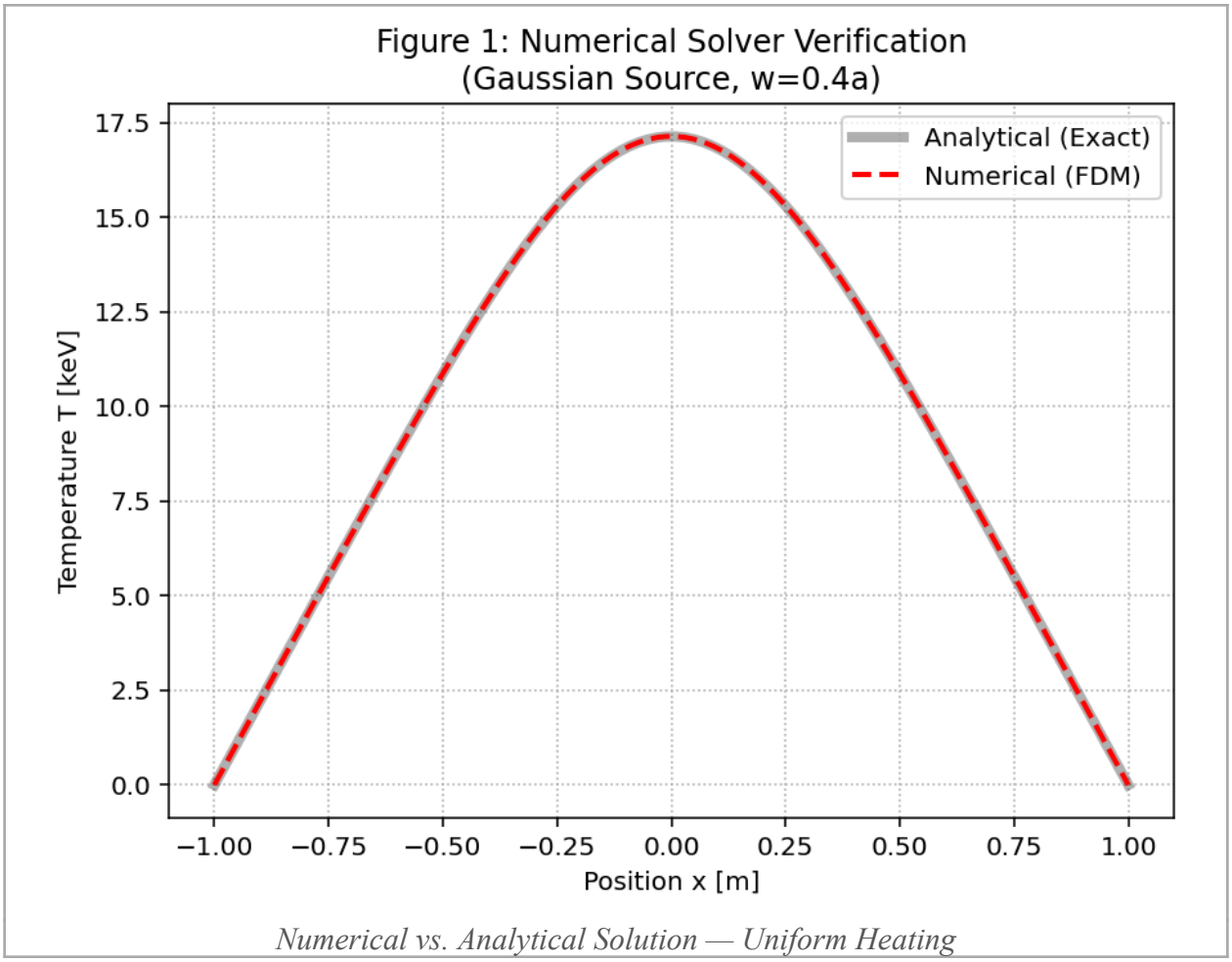


Figure 1. *Comparison of the numerical finite difference solution against the analytical exact solution (erf-based) for a Gaussian heating profile (w = 0.4a) at N = 100 grid points.*

2.4 Convergence Analysis

To rigorously establish the accuracy of the numerical scheme, a formal grid convergence study was performed. The solver was run for six successive grid resolutions: N = 10, 20, 40, 80, 160, and 320. At each resolution, the maximum absolute error between the numerical and analytical solutions was computed using the L∞ norm:

$$E(\Delta x) = \max_i \left| T_i^{num} - T_i^{analytical} \right|$$

Figure 2 presents the resulting error as a function of grid spacing Δx on a log-log plot, alongside a reference line of slope 2 representing the theoretical $O(\Delta x^2)$ scaling expected for a second-order accurate method. The measured error closely tracks this reference line across all grid sizes, confirming that the central difference scheme achieves second-order convergence. This result is consistent with the standard truncation error analysis of central differences, which yields a leading truncation error term of order $(\Delta x)^2 \, f^{(4)}(x)/12$ (Quintana-Kuether, 2026). Numerical noise does not distort the results because the problem involves a smooth, analytic solution with no sharp gradients.

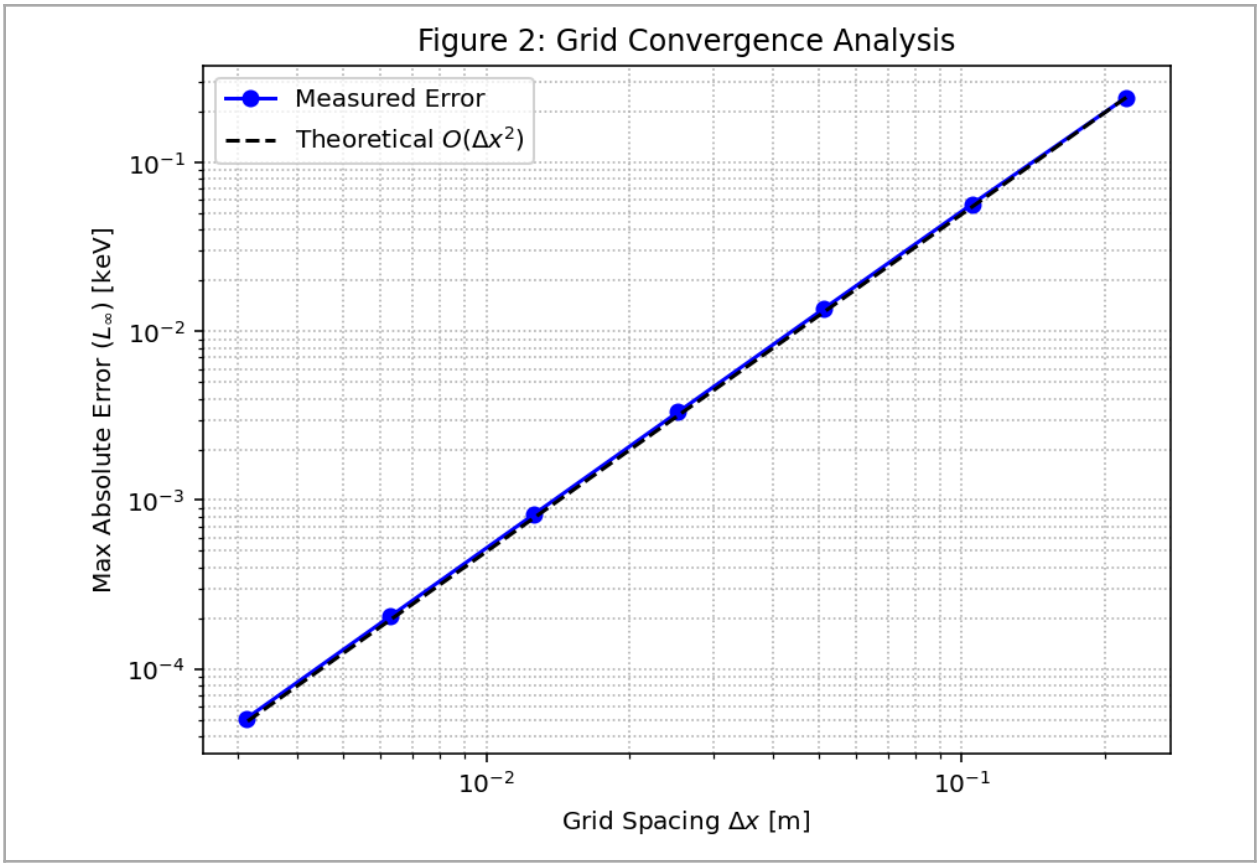


Figure 2. *Maximum absolute error between the numerical and analytical solutions as a function of grid spacing Δx, plotted on a log-log scale. The measured error (blue circles) aligns with the theoretical O(Δx²) reference line (dashed), confirming second-order accuracy of the central difference scheme.*

3. Results and Discussion

3.1 Baseline Verification and Context

With the solver verified, the baseline uniform-heating case yields a peak temperature of $T_0 \approx 31.2$ keV at the slab center. This result is physically meaningful: the ITER core plasma is designed to operate at an electron density of approximately 1×10^{20} m^{-3} and a central electron temperature of approximately 25 keV (Albajar et al., 2004). The model's baseline prediction of 31.2 keV is in the same order of magnitude as real device parameters, lending credibility to the choice of physical inputs and confirming that the simplified 1-D slab is capturing realistic energy balance physics, despite its geometric simplifications.

3.2 Sensitivity to Heating Profile Shape

The first sensitivity study examines how the spatial distribution of the heating source S(x) affects the steady-state temperature profile, with total deposited power held constant across all cases through normalization. Three heating distributions are compared: a uniform profile, a centrally peaked Gaussian (on-axis heating), and a double-Gaussian profile centered at x = ±0.5a (off-axis heating). Figure 3 presents the resulting temperature profiles.

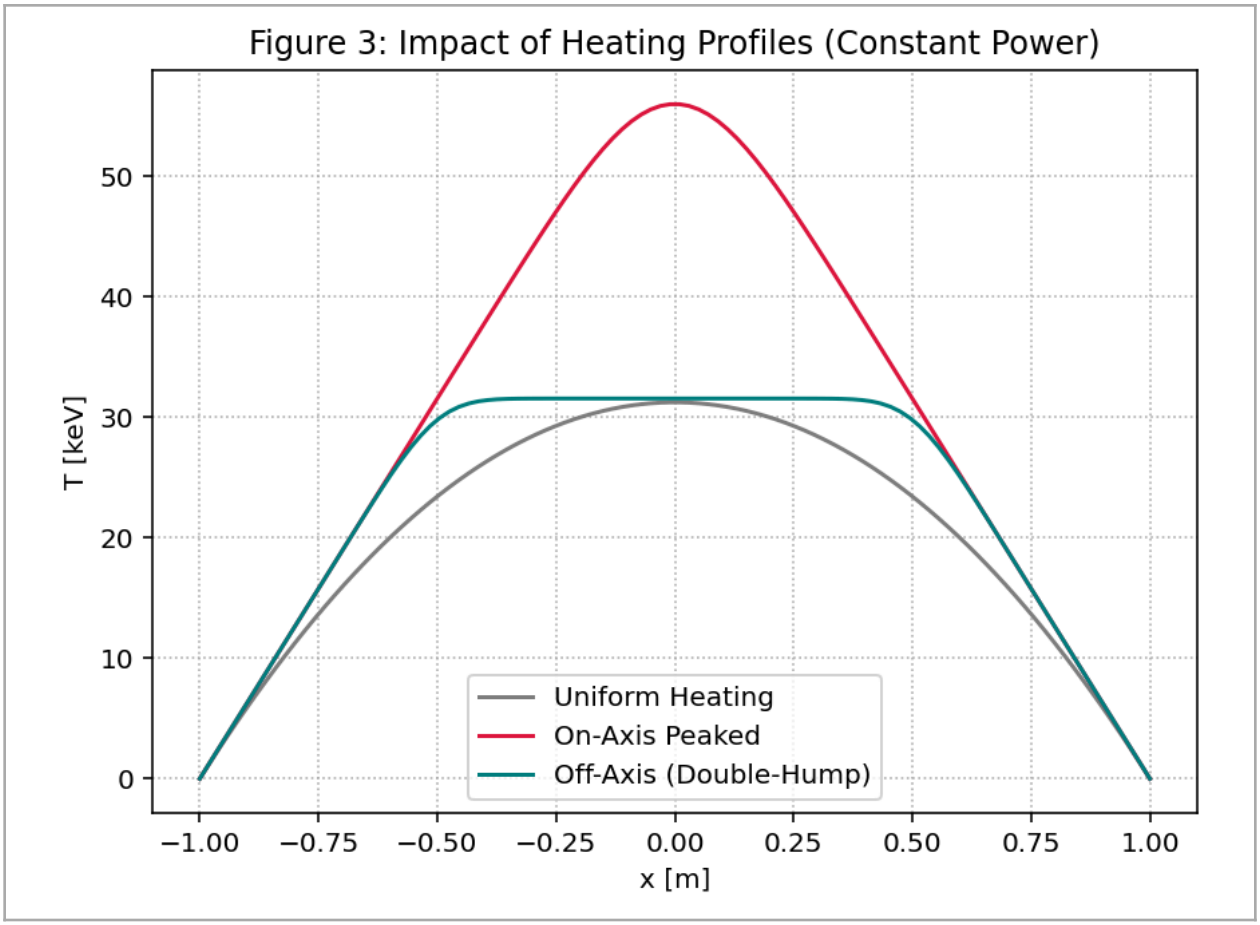


Figure 3. *Steady-state temperature profiles for three heating source distributions: uniform (gray), centrally peaked Gaussian (crimson), and off-axis double-Gaussian (teal). Total deposited power is held constant across all cases. The heating source profiles are reflected in the shape and peak of the temperature distribution.*

The on-axis peaked case produces the highest core temperature, while the uniform case yields the lowest. Table 1 summarizes the peak temperatures across all three cases.

Heating Profile	Peak Temperature (keV)
Uniform	31.21
On-Axis Peaked	55.93
Off-Axis (Double Hump)	31.53

Table 1. *Peak plasma temperatures for three heating profiles ($n = 1\times10^{20}$ m^{-3}, $\chi = 1.0$ m^2/s, $S_0 = 1$ MW/m^3).*

These results are physically intuitive. When heating is concentrated at the center, the local source term is large where the temperature gradient is smallest, requiring a high peak temperature to

drive sufficient heat flux to the boundaries. Conversely, off-axis heating deposits energy closer to the walls, which reduces the temperature gradient required to transport that energy outward and results in a flatter gradient. This behavior has direct engineering relevance: Neutral Beam Injection (NBI) in real tokamaks often deposits power off-axis due to geometric constraints, whereas electron cyclotron resonance heating (ECRH) can be focused on-axis. The model quantitatively captures the consequence of this distinction on energy confinement.

3.3 Sensitivity to Thermal Diffusivity

The second sensitivity study examines the role of the thermal diffusivity χ, which characterizes how readily thermal energy is transported across the plasma. Four values are tested: $\chi = 0.5$, 1.0, 2.0, and 4.0 m²/s, spanning a factor of eight. Uniform heating is used throughout. Figure 4 presents the temperature profiles for each case.

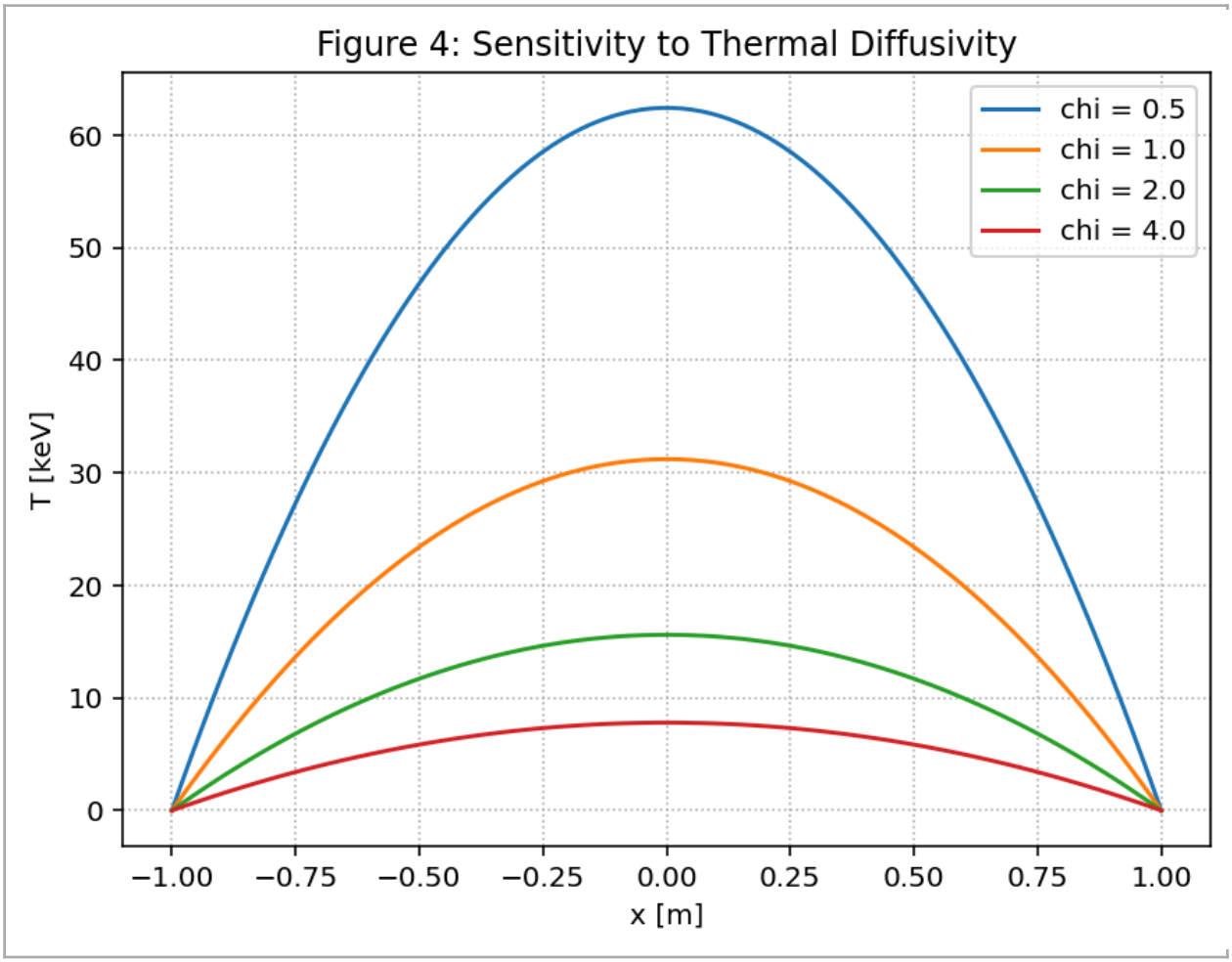


Figure 4. *Steady-state temperature profiles for four values of thermal diffusivity χ (0.5 to 4.0 m²/s) under uniform heating. Peak temperature decreases monotonically with increasing χ, consistent with the analytical inverse relationship $T_0 = S_0 a^2 / (2n\chi k_conv)$.*

Table 2 summarizes the peak temperature for each value of χ.

χ (m²/s)	Peak Temperature (keV)
0.5	62.42
1.0	31.21
2.0	15.61
4.0	7.80

Table 2. *Peak plasma temperatures as a function of thermal diffusivity χ, with uniform heating (S_0 = 1 MW/m³).*

The results confirm the analytically predicted inverse relationship: $T_0 \propto 1/\chi$. Doubling χ from 1.0 to 2.0 m²/s precisely halves the peak temperature from 31.2 to 15.6 keV, and halving χ from 1.0 to 0.5 m²/s exactly doubles it to 62.4 keV. This clean scaling is a direct consequence of the linearity of the governing equation and provides an additional implicit verification that the numerical solver is correct.

The physical implication is significant. In real tokamaks, anomalous transport driven by plasma turbulence causes χ to be considerably larger than classical neo-classical predictions, often by an order of magnitude (Wesson, 2004). The χ = 2.0 m²/s case yields a peak temperature of approximately 15.6 keV, closely matching ITER's core temperature design target of around 15–25 keV (Albajar et al., 2004). This alignment suggests that the simplified 1-D model, when parameterized with realistic transport coefficients, is capable of reproducing order-of-magnitude agreement with the temperature profiles observed in operating devices. By contrast, the χ = 4.0 m²/s case yields a peak of only 7.8 keV. This is well below the approximately 10 keV threshold generally considered necessary for deuterium-tritium ignition, illustrating how enhanced transport losses can prevent a reactor from reaching ignition conditions.

4. Conclusions

This project developed and verified a finite difference solver for the one-dimensional steady-state plasma energy balance equation, based on the physical model presented in Freidberg (2007). The central difference discretization was shown to be second-order accurate in grid spacing through a formal convergence study, and the solver was verified against the analytical exact solution for a Gaussian heating source. These verification steps establish that the solver is both correct and reliable.

The sensitivity studies revealed two key physical insights. First, the spatial distribution of the heating source meaningfully determines the peak plasma temperature even when total deposited power is held constant: on-axis peaked heating can raise the core temperature by approximately 79% relative to uniform heating, while off-axis deposition is marginally higher than uniform. This has practical relevance to the design and operation of auxiliary heating systems in tokamaks. Second, the peak temperature scales exactly inversely with thermal diffusivity, and the

range of χ values explored brackets the ITER design target — a result that lends physical credibility to the model despite its geometric simplifications.

The primary limitation of this model is its one-dimensional, steady-state, constant-coefficient formulation. Real plasma transport is fundamentally two or three-dimensional, time-dependent, and characterized by spatially varying and turbulently enhanced transport coefficients. Radiation losses and the coupling between ion and electron temperatures are also neglected. Potential extensions of this work include implementing a spatially varying $\chi(x)$ profile to model internal transport barriers, adding a radiation loss term to assess its effect on power balance, or extending the solver to two spatial dimensions using a sparse matrix approach. A time-dependent solver could also be developed to study transient plasma dynamics during heating ramp-up or disruption events.

References

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